Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/24/2024
PI Name:	Jain, Abhishek Ph.D.		
Project Title:	Long-term Patient iPSC Vessel Chip Model to Assess Stressors of Atherosclerosis and mRNA Therapeutics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	a.jain@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-458-8494
Organization Name:	Texas A&M Engineering Experiment Station		
PI Address 1:	101 Bizzell St, ETB 5045		
PI Address 2:	3120 TAMU		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	03/20/2023	**End Date:**	02/20/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Le, Nhat Tu Ph.D. (Houston Methodist Hospital) Yakovlev, Vladislav Ph.D. (Texas A&M Engineering Experiment Station) Wang, Guangyu Ph.D. (Houston Methodist Hospital) Cooke, John P M.D., Ph.D. (The Houston Methodist Research Institute)		
Grant/Contract No.:	80ARC023CA002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Animals are not very precise modeling systems of human physiology and cannot ensure dissectible signaling analysis. Recent advances in organ-on-chip technology or microphysiological systems (MPS) have permitted co-culture of human cells in physiologically-relevant microfluidic environments and also allowed continuous perfusion, providing an alternative in vitro approach to mimic the human disease. These microsystems, however, still have not reached their full potential and made in-depth entry into informing clinical trials due to the absence of efficient and scalable primary human patient cells, limited real-time biosensing potential, and critical validation over long durations. Amongst all human disease conditions, vascular diseases can be often fatal in which symptoms such as aggressive atherosclerosis exist. Amongst other risk factors, age is one of the most critical. Radiation exposure during space travel is another. Therefore, investigating vascular signaling mechanisms and finding therapeutics for aging and radiation-related vascular diseases is of high prospective impact. But there is a fundamental gap in the understanding of: 1) how organ-level signaling occurs in blood vessels amongst atherosclerotic patients, and 2) how cells inside the vascular organ contribute to the disease and respond to drugs over time. To meet these complex challenges, the team's goal is to take Progeria patients and proton radiation as case-studies and demonstrate that 3D cylindrical vessel-chips containing co-culture of vascular endothelial and smooth muscle cells from human induced pluripotent stem cells (hiPSCs) will provide real-time readouts of critical vascular and hematological biomarkers, that inform clinical outcomes. The chip will consist of physiologically-relevant extracellular matrix that will be continually monitored and it will also be ready for microgravity experiments that can be done in space. The objective is to establish long-term culture and feasibility of a Vessel-Chip system that helps dissect pathophysiology of atherosclerosis and toxin/drug-tissue interactions with much higher specificity and sensitivity than gold-standard models. We will add more advanced biosensing to our existing Vessel-Chip that mimics vessel geometry, hemodynamics, and extracellular matrix, to continuously monitor physiological parameters and sustain the culture of human iPSC-derived endothelial cells (hiPSC-EC) and vascular smooth muscle cells (hiPSC-SMC) for 180 days from healthy patients (Phase I). We will assess HGPS patient iPSC vascular cells and protons as biological and space-relevant stressors respectively over 180 days, and finally, novel Good Manufacturing Practice (GMP)-grade mRNA therapeutics loaded in a unique lipid nanoparticle-based delivery vehicle will be evaluated through a panoply of organ-to-gene level assays, including clinically-relevant biomarkers (Phase II). The research proposed is innovative and significant, because it contributes in shifting the status quo of vascular pathophysiology in the organ-on-a-chip platform by containing primary hiPSC-vascular cells, blood flow, and associated transcriptomics and proteomics over long durations of co-culture and continuous monitoring. The outcomes will lead to further applications of the organ-on-chip technology in translational research in atherosclerosis and other vascular and hematological diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: -- A long-duration microphysiological model of atherosclerosis. -- New knowledge of aging and radiation-induced vascular disease. -- A testbed for mRNA therapeutic preclinical trials. -- Enabling future experimentation at the International Space Station (ISS) under microgravity.
	Our overall objective in Phase 1 is to engineer a vessel-chip that is cultured for a prolonged period of time (up to 180 days). To achieve the objective, our focus has been to optimize the hemodynamic conditions, cell culture media factors, environmental stability, and analytical strategy. The state of the art of vessel-chips is currently a median level of 4 days and a maximum of 14 days of living culture. However, in year 1, we have achieved a sustainable culture of a vessel-chip consisting of living endothelial cells for 60 days under flow. Our specific outcomes are as follows: 1. We have identified strategies and developed robust protocols to obtain human induced pluripotent stem cells (hiPSCs) from parents of Hutchinson-Gilford Progeria Syndrome (HGPS) parents, and subsequently differentiate these stem cells into endothelial cells (ECs) and vascular smooth muscle cells (VSMCs). We optimized the growth factor cocktail that's added to the culture media and optimized the substrate Extracellular matrix (ECM) coating. 2. We can seed induced pluripotent stem cell derived endothelial cells (iPSC-ECs), human umbilical vein endothelial cells (HUVECs) and human aortic endothelial cells (HAECs) in vessel-chips, and keep them perfused under a constant flow of culture-media. Utilizing this approach, we have been able to sustain endothelial culture in vessel-chips for up to 60 days, with ECs expressing stable vascular-adherens junctions at the 60-day mark, as investigated by immunofluorescent staining and microscopy. In addition, we can obtain live time-lapse images of 2 devices at a time to monitor and assess the changes in endothelial health during the culture period. 3. We have obtained preliminary data from an automated morphology-based culture analysis pipeline using the AI tool Cellpose and analysis tool CellProfiler. This tool is expected to assist with the quantification of cell health and remove human bias from measurements. 4. Several technical challenges have been addressed. We had to optimize the fluidic circuit for long-term culture to minimize the flow-perturbations during media changes, which is usually not required for short-term experiments. We had to optimize the chemistry of ECM tethering to the polydimethylsiloxane (PDMS) device by including pre-treatment steps; otherwise, we noticed that the ECM-lumen tends to collapse within 5-7 days of culture. Further modifications are required in the pumping apparatus because the mechanical components are wearing out over time due to friction. 5. We tested the performance of commercial syringe and peristaltic pumps by prescribing uniform flow to the pumps and measuring the resulting flow profile using a high data acquisition (DAQ)-rate calorimetric flow sensor, and subsequently analyzing the noise in flow utilizing Fast Fourier Transform analysis and measuring the signal to noise ratio. We found that the peristaltic pump had the highest noise in flow-rate, and characteristic retrograde flow pulses were identified. The syringe pump had a relatively stable flow profile, however it had antegrade flow pulses, which could also be detrimental to endothelial health, especially in long-term culture, albeit not as much as retrograde pulses. 6. We have engineered a new perfusion technology, Hemadyne, for mimicking arterial perfusion at physiological time-scales for long-term culture, and the pump was fabricated using 3D printing. This pump works on the principle of pneumatically induced flow and can recapitulate human in vivo hemodynamic flow-patterns in an in vitro vessel-chip system. We found that for constant flow, Hemadyne had a notably higher flow stability than a peristaltic and syringe pump. We recapitulated physiological and pathological multiphasic waveforms and found that the pump was able to recapitulate these waveforms with high fidelity. 7. We systematically tested various conditions known to influence cellular longevity, assessing the impact of specific

Task Progress:	molecules and environmental factors. These included factors known to enhance cell survival pathways, improve metabolic signaling, reduce oxidative stress, enhance angiogenesis, and augment homeostatic mechanisms known to maintain endothelial viability and identity. The following conditions were tested in 2D culture using standard endothelial growth medium, plus the following enhancements: a. VEGF (10 and 30 ng/ml): Contributes to cell longevity by promoting EC functions and survival. b. FBS (1% and 3%): Reduced FBS promotes longevity. c. H2S (0.3 and 1 mM): Enhances cellular longevity by reducing oxidative stress and inflammation, and improving mitochondrial function. d. NAD precursor (0.3 and 1 mM): Enhances longevity by boosting cellular NAD+ levels, which decline with age, supporting energy metabolism and DNA repair. e. Temperature at 35°C: Increases longevity by reducing metabolic rate and cellular stress. f. Telomerase: Reduces senescent properties and maintains mitochondrial and telomere function. g. 8-Br-cAMP (100 and 300 mM/ml): Acts as a "second messenger" for the EC survival factor prostacyclin. It activates protein kinase A (PKA) and modulates various cellular pathways related to stress resistance, energy metabolism, and apoptosis. h. 8-Br-cGMP (100 and 300 mM/ml): Serves as a "second messenger" for the EC survival factor nitric oxide. It extends cell lifespan in culture by activating cGMP-dependent pathways, which promote cell growth and survival, reduce inflammation, and enhance the cell's ability to respond to stress and damage. i. CHIR 99021 (10 and 30 ng/ml): An inhibitor of glycogen synthase kinase-3 (GSK-3). Protects cells from stress-induced damage, thereby contributing to an extended cell lifespan in culture. j. SB 431542 (10 and 30 µM): An inhibitor of TGF-ß. Increases cellular longevity in culture by reducing fibrosis and epithelial-mesenchymal transition (EMT), processes often associated with aging and cellular senescence. Growing human aortic endothelial cells (HAEC) and using cultured media modified with the above conditions, our preliminary data have indicated that in the first 2 weeks, little difference was observed in cell proliferation or morphology. However, in the ensuing weeks up to 40 days, several conditions clearly performed better in terms of EC viability and/or morphology. The winning conditions were reduced FBS, VEGF, and SB. Subsequently, we repeated these studies focusing on the winning conditions. In addition, we assessed the combination of SB and VEGF, and we assessed lower concentrations of cAMP and cGMP. UMAP, GO and heatmap analyses of the various conditions suggested that VEGF treatment was superior to the other conditions in providing a long-term maintenance of endothelial viability and identity. However, there were significant transcriptional differences in these cells with some activation of inflammatory and neuronal pathways. 8. Mycoplasma Control: One of the most common and significant challenges is contamination by bacteria, fungi, and mycoplasma. Contaminants can alter cell physiology, affect experimental outcomes, and may not always be immediately visible, leading to prolonged issues before detection. We have developed several state-of-the-art strategies to counteract this problem. 9. We have optimized the conditions and are now growing cells on the vessel-chip at the Houston Methodist Research Institute (HMRI). Although we have not yet used a flow pump to induce flow, the optimized conditions appear to be aiding cellular longevity based on cellular morphology. We are collecting media at different timepoints from the vessel-chip for multiplex enzyme-linked immunosorbent assay (ELISA) analysis to identify patterns of cytokine expression over time. Additionally, we are sacrificing vessel-chips at different timepoints for further analysis of senescent properties. 10. We have successfully identified a protocol of obtaining high-purity bulk RNA from endothelialized vessel-chips for subsequent transcriptomic analysis. We get about 230 ng RNA from each vessel-chip, which is enough for most transcriptomic analysis pipelines.
Bibliography Type:	Description: (Last Updated: 06/02/2026)
Abstracts for Journals and Proceedings	Kumar A, Patel RD, Jain,A. "Hemadyne: Ultrafast pump recreates clinical hemodynamics and endothelial responses in preclinical microphysiological systems." 8th Annual Cardiovascular Bioengineering Symposium, Houston, TX, May 22-25, 2024. Abstracts. 8th Annual Cardiovascular Bioengineering Symposium, Houston, TX, May 22-25, 2024. , May-2024
Abstracts for Journals and Proceedings	Lee J, Mathur T, Kumar A, Jain A. "Gravitational lumen patterning of complex vascular architectures: Fabricating collagen embedded lumen for modeling spatially intricate blood vessels." 5th Annual Gulf Coast Vascular Research Consortium, Houston, TX, August 4, 2023. Abstracts. 5th Annual Gulf Coast Vascular Research Consortium, Houston, TX, August 4, 2023. , Aug-2023
Articles in Peer-reviewed Journals	Ghosh LD, Mathur T, Tronolone JJ, Chuong A, Rangel K, Corvigno S, Sood AK, Jain A. "Angiogenesis-enabled human ovarian tumor microenvironment-chip evaluates pathophysiology of platelets in microcirculation." Adv Healthc Mater. 2024 Mar 30;304263. https://doi.org/10.1002/adhm.202304263 ; 38553940; PMCID: PMC11281868 , Mar-2024
Articles in Peer-reviewed Journals	Deo KA, Murali A, Tronolone JJ, Mandrona C, Lee HP, Rajput S, Hargett S, Selahi A, Sun Y, Alge D, Jain A, Gaharwar AK. "Granular biphasic colloidal hydrogels for 3D bioprinting." Adv Healthc Mater. 2024 May 15:e2303810. https://doi.org/10.1002/adhm.202303810 ; PMID: 38749006 , May-2024